

Solution Architecture

Team ID: SWTID-2026-4623

Team Size:5

Project Name: Import Data using Transform Maps (Spreadsheet)

Solution Architecture Diagram

Presentation / Client Layer

ServiceNow Web Interface

- Import Set / Load Data interface
- Transform Map configuration
- Employee records
- Reports and Employee Analytics Dashboard

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API Gateway / Load Balancer

ServiceNow Platform Request Layer

- Handles user requests within the ServiceNow instance
- Connects the ServiceNow interface with import and transformation processes

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Auth Service

ServiceNow Authentication & Role-Based Access

- Authenticates ServiceNow users
- Controls access based on assigned roles and permissions
- HR/System Administrators can perform authorized import and monitoring activities

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Core Logic Service

ServiceNow Import Sets & Transform Maps

- Loads spreadsheet data into the Import Set staging table
- Maps source fields to target fields
- Transforms employee information
- Uses Coalesce on Employee ID to identify existing records

- Inserts new records and updates existing records

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External API / Integrations

Excel / Google Sheets

- External source for structured employee data
- Provides Employee ID, Employee Name, Email, Department, and Location
- Data is prepared before being uploaded to ServiceNow

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Data / Storage Layer

ServiceNow Database

- Employee Import staging table
- Employee Test (u_employee_test) target table
- Transform history and import results
- Reports and dashboard data

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the interface for uploading employee spreadsheets, configuring Transform Maps, viewing employee records, reports, and dashboards.	ServiceNow Web Interface
API Gateway / Request Layer	Handles requests between the ServiceNow user interface and platform services during import and transformation activities.	ServiceNow Platform
Authentication / Authorization	Authenticates users and controls access to employee data and import functionality using roles and permissions.	ServiceNow Authentication & Role-Based Access
Core Logic Service	Processes imported spreadsheet data, performs field mapping and	Import Sets, Transform Maps,

Component Name	Description / Role in Architecture	Technologies Used
	transformation, applies Coalesce, and inserts or updates employee records.	Mapping Assist, Coalesce
External Data Source	Provides structured employee information that is uploaded into ServiceNow for processing.	Microsoft Excel / Google Sheets
Database / Data Layer	Stores imported staging data, transformed employee records, and import/transform results.	ServiceNow Database
Reporting & Dashboard	Displays employee information and analytics such as department, location, employee list, and import results.	ServiceNow Reports & Dashboards

Overall Architecture Flow

Excel / Google Sheets



ServiceNow Web Interface



Import Set (Employee Import)



Transform Map + Coalesce



Employee Test Table (u_employee_test)



Reports & Employee Analytics Dashboard

Architecture Summary

The architecture uses ServiceNow as the central application platform. External spreadsheet data enters through the ServiceNow Web Interface, is temporarily stored in an Import Set, and is processed using Transform Maps. Coalesce using Employee ID prevents duplicate records by updating existing employees. The processed data is stored in the ServiceNow database and presented through reports and an Employee Analytics Dashboard.